

HYBRID INTRUSION PREVENTION SYSTEM AND MACHINE LEARNING

**MULTI STAGE THREAT DETECTION IMPLEMENTATION
FOR IDENTIFYING UNREGISTERED CYBER ATTACKS**



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01

Escalating Cyber Threats in Indonesia

According to the Indonesia Cybersecurity Landscape 2024 report, total detected anomaly traffic reached 330,527,636 throughout the year.

03

Limitations of Current Systems

Existing Intrusion Prevention Systems (IPS) still heavily rely on signature-based detection that can only identify known attacks in database.

RESEARCH BACKGROUND

02

High Volume of Anomaly Traffic

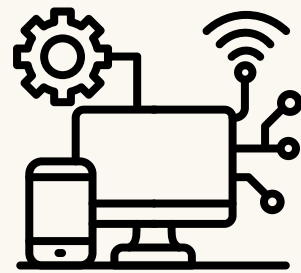
Indonesia faces significant cyber risks as both a source and target:

- Source: 164,432,759 anomalies.
- Destination: 194,562,390 anomalies.

PROBLEM ANALYSIS

1 Technological Aspect

Existing security frameworks (IDPS, SIEM, IAM/MFA) struggle to detect zero-day attacks and dynamically evolving threats.



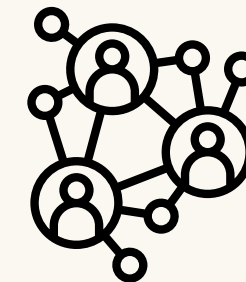
2 Security & Cyber Threat Aspect

BSSN (2024) reported over 330.5 million anomaly cases and 514,508 ransomware activities, highlighting a critical security crisis.



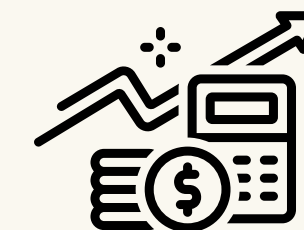
3 Network & Infrastructure Aspect

High internet penetration (221.5 million users) and infrastructure complexity have created numerous security gaps and real-time response limitations.

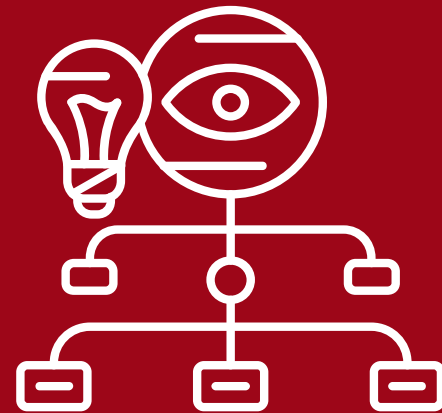


4 Economic Aspect

Cyberattacks cause massive financial damage, with regional losses in West Java reaching 254 billion IDR per hour.



GOALS



01

**Developing ML-based
IPS to adaptively
detect new attacks.**

**Reducing false
positives in security
systems with more
accurate ML models.**

02



03

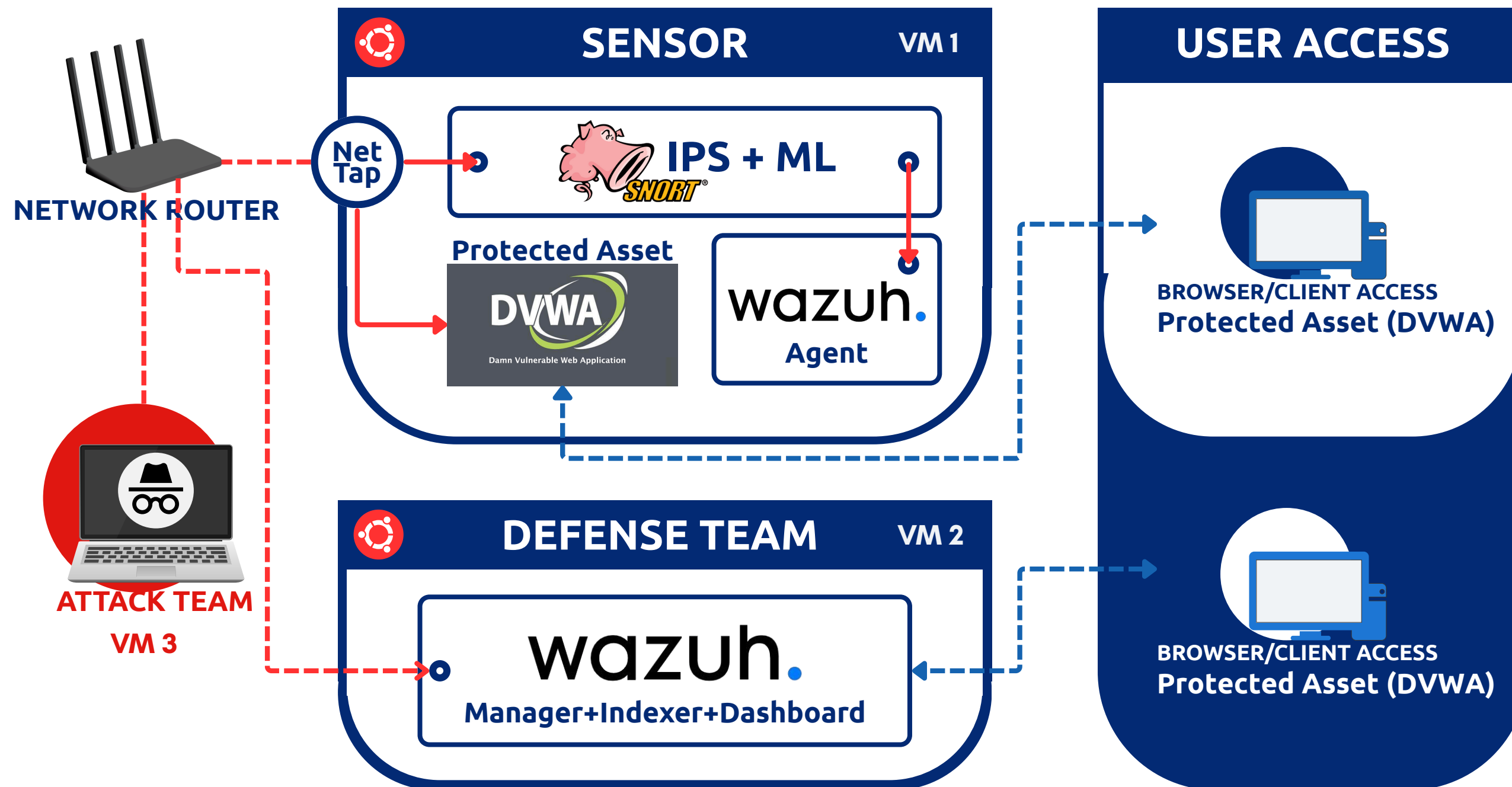
**Improving the efficiency
of cyber threat
detection and response
while maintaining
accuracy.**



**Universitas
Telkom**

SOLUTIONS

HOW DOES THE SYSTEM WORK?



OVERALL SYSTEM INTEGRATION

DEPLOYMENT OF SNORTML IPS WITH DAQ AFPACKET AND NFQ MECHANISMS

DAQ AFPACKET

```
snort@snort:/usr/local/etc/snort/log$ sudo snort -c /usr/local/etc/snort/snort.lua --talos --lua 'snort_ml_engine = { http_param_model = "/usr/local/etc/snort/models/ae_classifier_model.tflite" }; snort_ml = { client_body_depth = -1, http_param_threshold = 0.95 }; trace = { modules = { snort_ml = { all = 1 } } };' -A alert_fast -l /var/log/snort -s 65535 -k none -Q --daq afpacket -i enp0s3
```

DAQ NFQ

```
snort@snort:~$ sudo iptables -I INPUT -p tcp --dport 80 -j NFQUEUE --queue-num 0  
[sudo] password for snort:  
snort@snort:~$ sudo snort -c /usr/local/etc/snort/snort.lua --talos --lua 'snort_ml_engine = { http_param_model = "/usr/local/etc/snort/models/ae_classifier_model.tflite" }; snort_ml = { client_body_depth = -1, http_param_threshold = 0.95 }; trace = { modules = { snort_ml = { all = 1 } } };' -A alert_fast -l /var/log/snort -s 65535 -k none -Q --daq nfq --daq-var queue=0
```

ATTACKING SIMULATION

```
query): tWfNbk=Al1NTJ8sy6jU  
1.000000  
query): U171nT=EbYQSn6cC&CwBc=Kgqur20RWkfH8  
1.000000  
query): gRU4Qy7x=5KxjxCt7d&DiNxPG=VOa5FhN3g  
1.000000  
query): M5JxGW=Y85Bes&wToYkY4sg=gvl0HIA  
1.000000  
query): LQh7=GxRBudGi&GNTUfxKnr3=gQxFbDfIrv  
1.000000  
query): uYPcD7Jnt=TJalEicotLWf&PrXOch=MYfPj  
1.000000
```

```
wazuhwazuhmanager: ~$  
kali@kali: ~/GoldenEye  
'\_'  
| $$ \_/_/ /$$$$$ | $$ /$$$$$  
/$$$$$  
GoldenEye v2.1 by Jan Seidl <jseidl  
Hitting webserver in mode 'get' wit  
Hit CTRL+C to cancel.  
^CProcess Striker 0.  
CTRL+C received. Killing all worker  
0 GoldenEye strikes hit. (3539 Fail  
0 GoldenEye strikes hit. (3539 Fail  
0 GoldenEye strikes hit. (3539 Fail  
0 GoldenEye strikes hit. (3539 Fail  
0 GoldenEye strikes hit. (3539 Fail  
Shutting down GoldenEye  
(kali@kali)-[~/GoldenEye]  
$
```

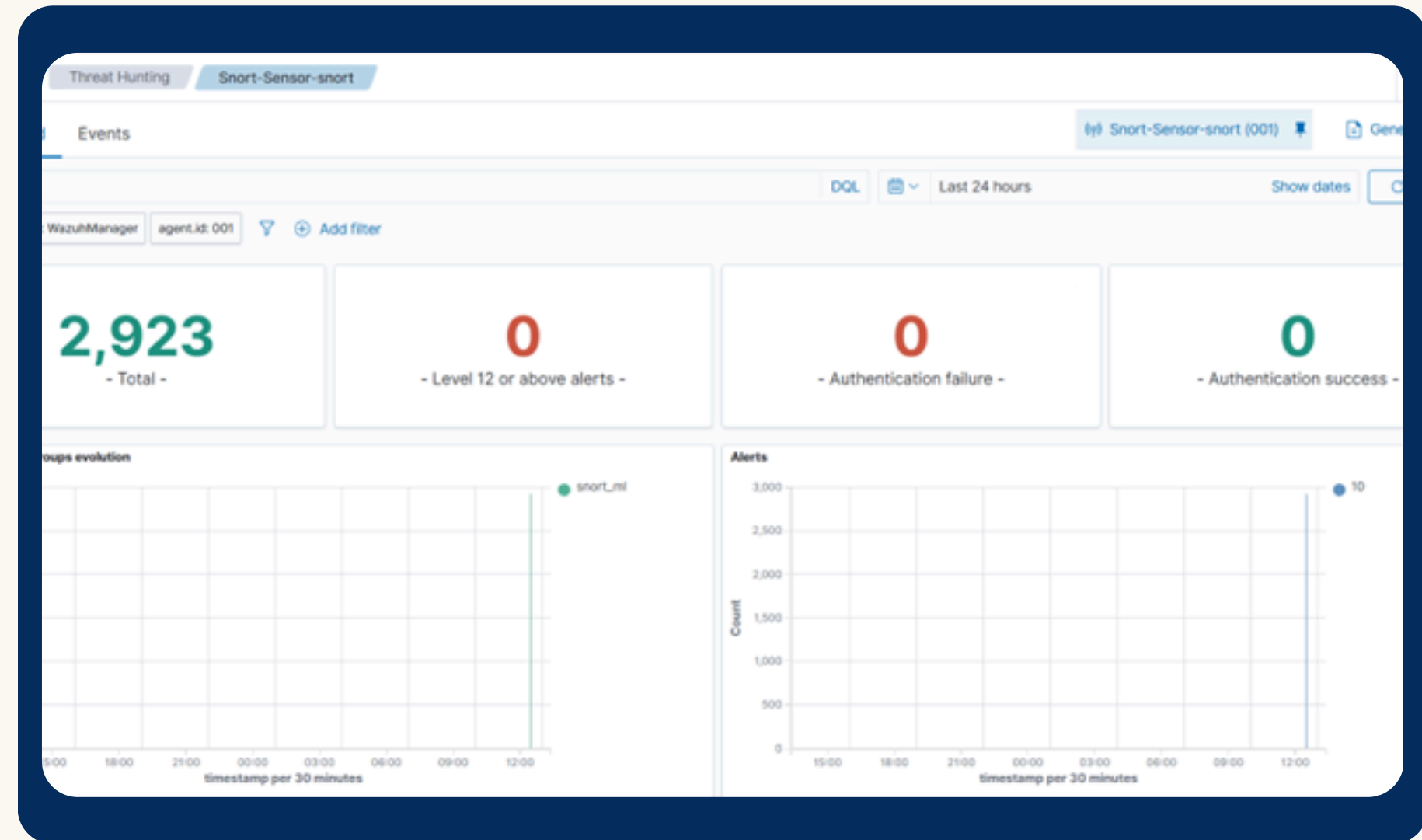
OVERALL SYSTEM INTEGRATION

LOGGING FILE SNORTML

```
:~/snort$ sudo tail -5 /var/log/snort/alert_fast.txt
:14.652598 [**] [411:1:1] <enp0s3> "(snort_ml) potenti
TP parameters via Neural Network Based Exploit Detection
{TCP} 192.168.18.24:41812 -> 192.168.18.21:80
:20.094975 [**] [411:1:1] <enp0s3> "(snort_ml) potenti
TP parameters via Neural Network Based Exploit Detection
{TCP} 192.168.18.24:60682 -> 192.168.18.21:80
:36.658233 [reset] [**] [112:1:1] <enp0s3> "(arp_spoof
" [**] [Priority: 3] {ARP} ->
:39.585117 [reset] [**] [411:1:1] <enp0s3> "(snort_ml)
nd in HTTP parameters via Neural Network Based Exploit
rity: 3] {TCP} 192.168.18.24:44490 -> 192.168.18.21:80
:44.552142 [reset] [**] [411:1:1] <enp0s3> "(snort_ml)
nd in HTTP parameters via Neural Network Based Exploit
rity: 3] {TCP} 192.168.18.24:44506 -> 192.168.18.21:80
```

03

WAZUH LOGGING



04

SOLUTIONS

HOW ABOUT MACHINE LEARNING?

BASIS FOR MODEL DETERMINATION:

Anomaly Attack Detection

01

Scalability

02

Adaptability

03

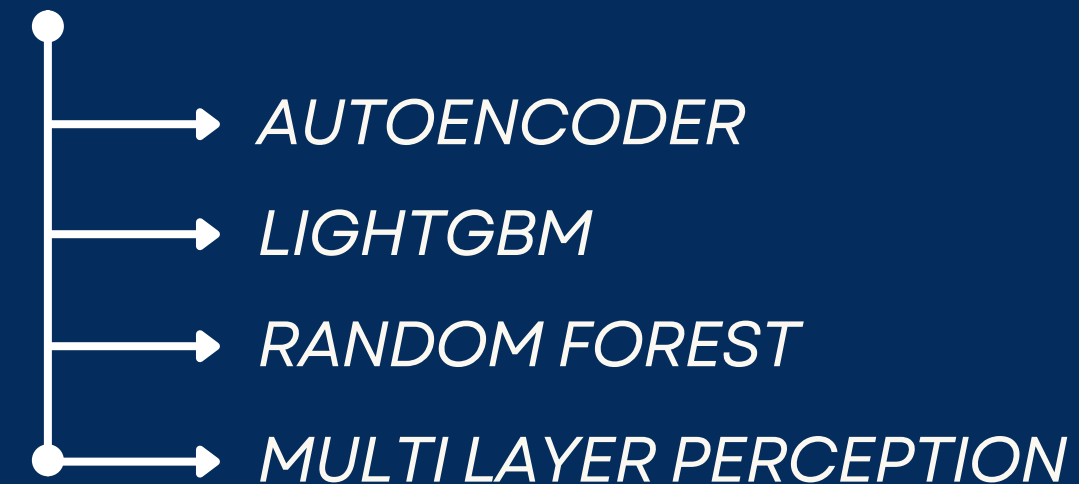
Complexity of Implementation

04

Previous Research

05

THEN, WHAT MODEL IS USED?

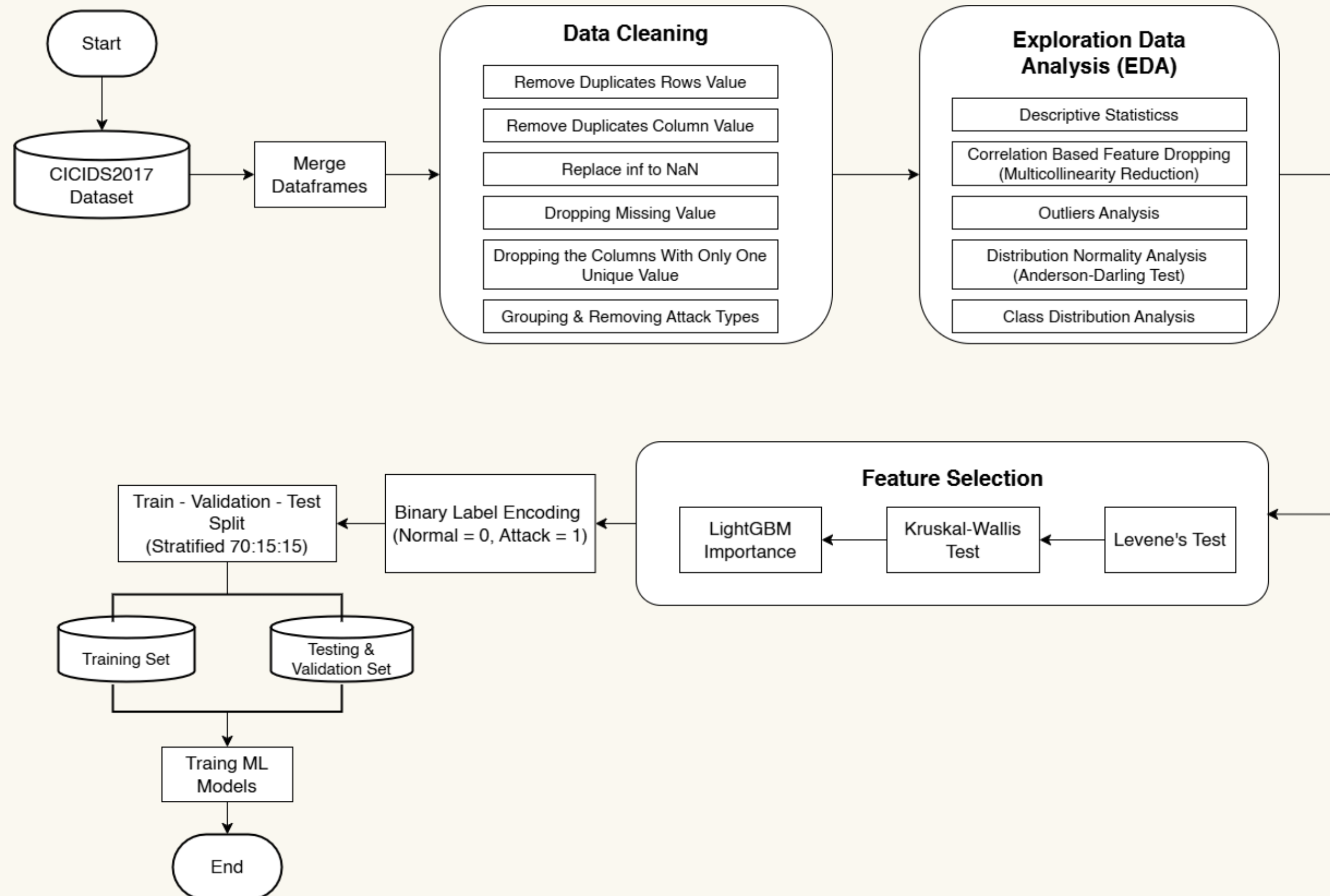


WHAT ABOUT THE DATASET?

CICIDS2017

- Spans from Monday through Friday
- Totaling roughly 2.8 million data points
- Encompassing 14 different attack types

DATA PRE-PROCESSING



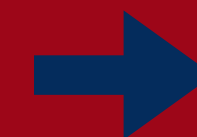
PRE-PROCESSING RESULT

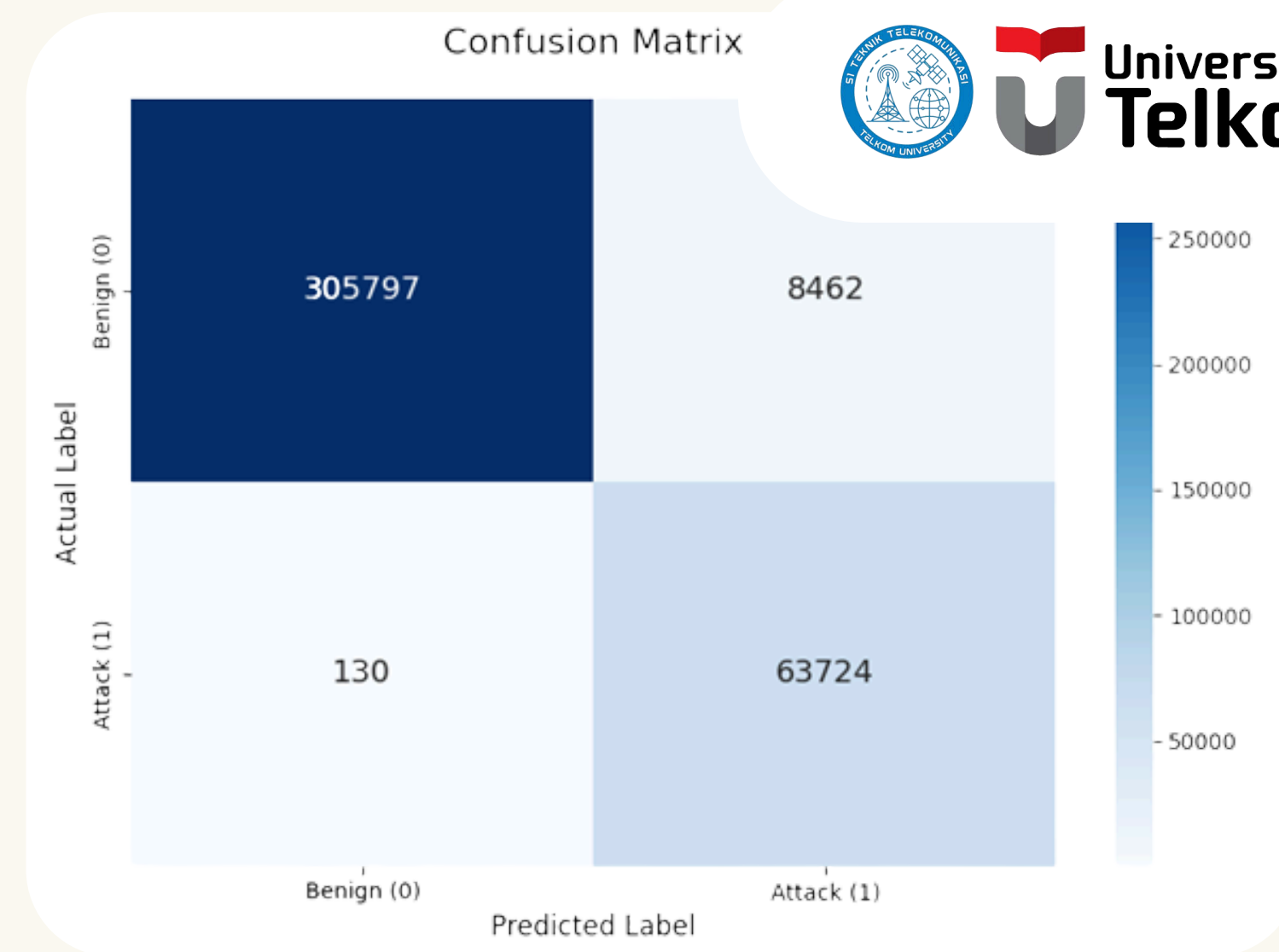
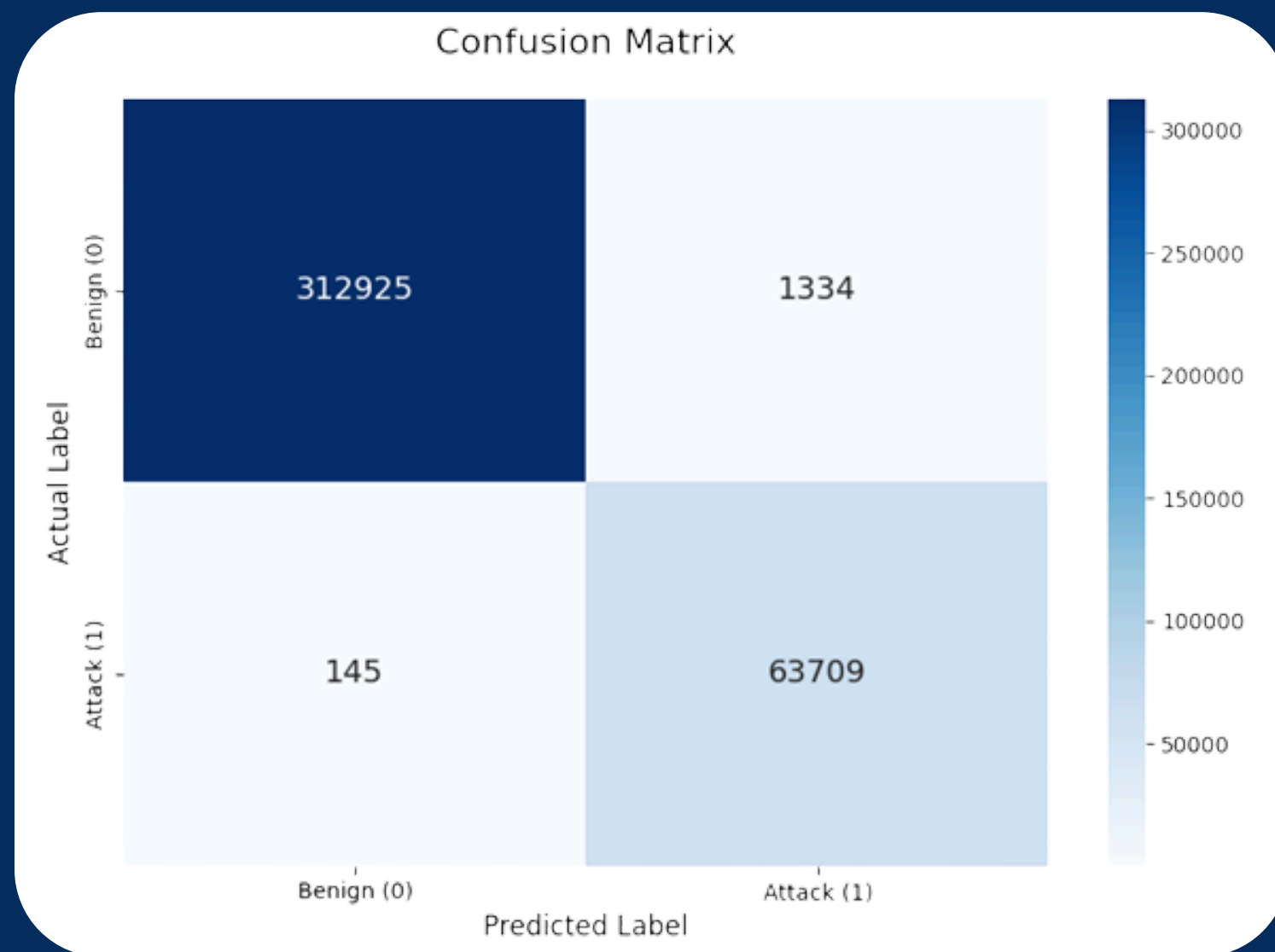
Final Dimention:
(2520751, 44)

Splitting Information

Training Data : (1764525, 43)
Validation : (378113, 43)
Testing Data : (378113, 43)

What are the results of
training the model using
this preprocessing?



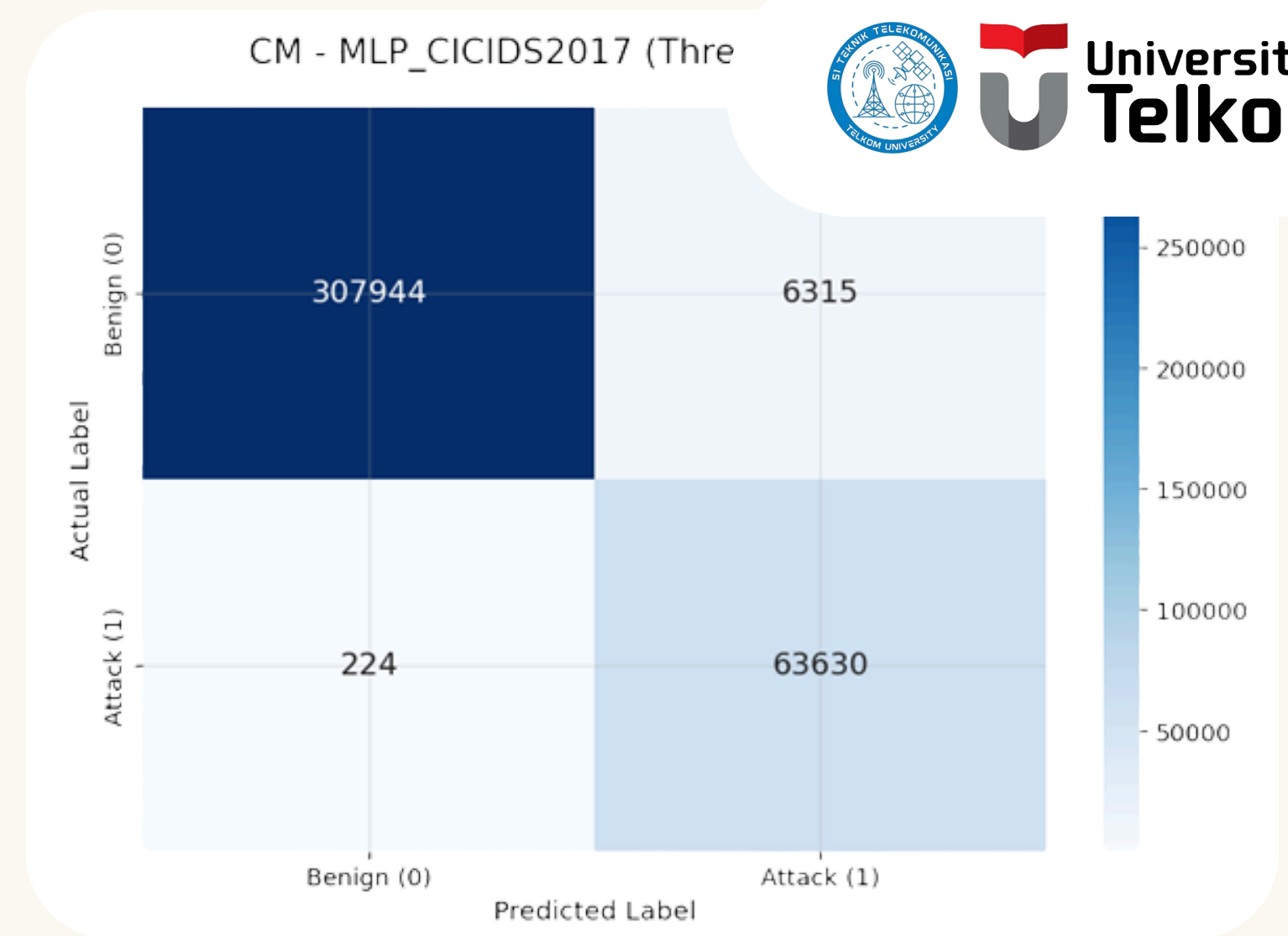
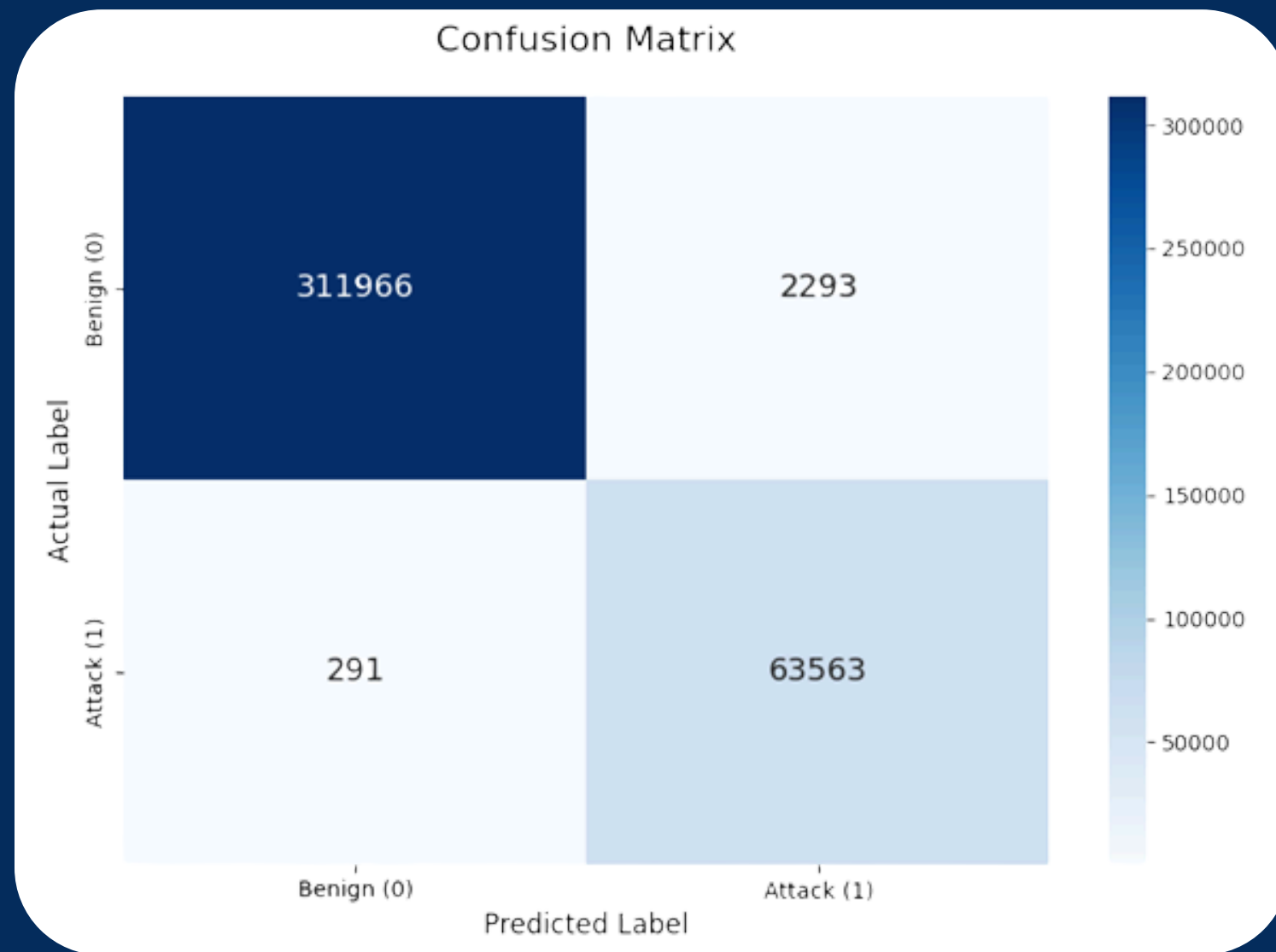


AUTOENCODER

Label	Precision	Recall	F1-Score	Support
Benign (0)	0,9995	0,9965	0,998	314.259
Attack (1)	0,983	0,9975	0,992	63.854
Accuracy			0,9967	378.113
Macro Avg	0,9912	0,997	0,9941	378.113
Weighted Avg	0,9967	0,9967	0,9967	378.113

LIGHTGBM

Label	Precision	Recall	F1-Score	Support
Benign (0)	0,9996	0,9731	0,9861	314.259
Attack (1)	0,8828	0,998	0,9368	63.854
Accuracy			0,9773	378.113
Macro Avg	0,9412	0,9855	0,9615	378.113
Weighted Avg	0,9799	0,9773	0,9778	378.113



RANDOM FOREST

Label	Precision	Recall	F1-Score	Support
Benign (0)	0,9983	0,9953	0,9968	314.259
Attack (1)	0,9775	0,9918	0,9946	63.854
Accuracy			0,9947	378.113
Macro Avg	0,9879	0,9936	0,9907	378.113
Weighted Avg	0,9948	0,9947	0,9947	378.113

MLP

Label	Precision	Recall	F1-Score	Support
Benign (0)	1	0,98	0,99	314.259
Attack (1)	0,91	1	0,95	63.854
Accuracy			0,98	378.113
Macro Avg	0,95	0,99	0,97	378.113
Weighted Avg	0,98	0,98	0,98	378.113

PERFORMANCE OF THE TRAINED MODEL



AUTOENCODER

Label	Precision	Recall	F1-Score	Support
Benign (0)	0,9995	0,9965	0,998	314.259
Attack (1)	0,983	0,9975	0,992	63.854
Accuracy			0,9967	378.113
Macro Avg	0,9912	0,997	0,9941	378.113
Weighted Avg	0,9967	0,9967	0,9967	378.113

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Benign (0)	1	0,98	0,99	314.259
Attack (1)	0,91	1	0,95	63.854
Accuracy			0,98	378.113
Macro Avg	0,95	0,99	0,97	378.113
Weighted Avg	0,98	0,98	0,98	378.113

MACHINE LEARNING ANALYSIS

AUTOENCODER

	Benign	Goldeneye	Slowloris	Brute Force	SQLi	XSS
Total Serangan	237	1329	650	81	128	4
Serangan Terdeteksi	0	1298	650	73	119	2
Persentase	0%	97,67%	100%	90,12%	92,97%	50%

Parameter	Nilai
FP	50
FN	0
TN	237
TP	2142
Recall	100%
Precision	97,72%
Accuracy	99,03%
F1-Score	98,85%

LIGHTGBM

	Benign	Goldeneye	Slowloris	Brute Force	SQLi	XSS
Total Serangan	237	1329	650	81	128	4
Serangan Terdeteksi	0	1092	0	73	0	2
Persentase	0%	82,17%	0%	90,12%	0%	50%

Parameter	Nilai
FP	1082
FN	0
TN	237
TP	1110
Recall	100%
Precision	50,64%
Accuracy	55,45%
F1-Score	67,24%

MACHINE LEARNING ANALYSIS

RANDOM FOREST

	Benign	Goldeneye	Slowloris	Brute Force	SQLi	XSS
Total Serangan	237	1329	650	81	128	4
Serangan Terdeteksi	0	1092	0	73	0	2
Persentase	0%	82,17%	0%	90,12%	0%	50%

Parameter	Nilai
FP	1052
FN	0
TN	237
TP	1167
Recall	100%
Precision	53,23%
Accuracy	57,80%
F1-Score	69,46%

MLP

	Benign	Goldeneye	Slowloris	Brute Force	SQLi	XSS
Total Serangan	237	1329	650	81	128	4
Serangan Terdeteksi	0	1092	649	73	0	2
Persentase	0,00%	82,17%	99,85%	90,12%	0,00	50%

Parameter	Nilai
FP	376
FN	0
TN	237
TP	1816
Recall	100%
Precision	82,85%
Accuracy	82,85%
F1-Score	90,63%

MACHINE LEARNING ANALYSIS

	Recall	Precision	Accuracy	F1-Score
Autoencoder	100%	97,72%	99,03%	98,85%
LightGBM	100%	50,64%	55,45%	67,24%
Random Forest	100%	53,23%	57,80%	69,46%
Multi Layer Perceptron	100%	82,85%	82,85%	90,63%

Skenario	cpu_mean	cpu_max	ram_mean	ram_max
SnortML IPS	7,69	32,43	3220,17	3281,93
Tanpa Snort	9,33	42,45	2972,56	3040,23
Selisih Perubahan (%)	-17,578	-23,604	8,33	7,95
BENIGN	disk_mean	disk_max	net_mean	net_max
	1119,78	19237,1	1,99	9,36
	5120,18	118798	68,01	993,24
	-78,13	-83,807	-97,074	-99,058

Skenario	cpu_mean	cpu_max	ram_mean	ram_max
SnortML IPS	29,67	73,71	3245,82	3291,18
Tanpa Snort	4,88	26,04	3006,11	3100,47
Selisih Perubahan (%)	507,99	183,06	7,97	6,15
DoS GOLDENEYE	disk_mean	disk_max	net_mean	net_max
	1526,93	17464,1	282,46	2296,51
	349,31	12701,5	106,12	1319,33
	337,13	37,5	166,17	74,07

Skenario	cpu_mean	cpu_max	ram_mean	ram_max
SnortML IPS	24,79	47,23	3263,26	3290,97
Tanpa Snort	4,37	18,64	3014,19	3075,16
Selisih Perubahan (%)	467,28	153,38	8,26	7,02
SLOW HTTP DoS	disk_mean	disk_max	net_mean	net_max
	477,76	16616,5	57,11	523,7
	80,8	1144,88	15,02	279,71
	491,29	1351,37	280,23	87,23

HARDWARE ANALYSIS

Skenario	cpu_mean	cpu_max	ram_mean	ram_max
SnortML IPS	23,44	29,3	3265,82	3283,61
Tanpa Snort	1,26	2,17	2997,92	3003,58
Selisih Perubahan (%)	1760,32	1250,23	8,94	9,32
BRUTE FORCE	disk_mean	disk_max	net_mean	net_max
	218,29	22003,78	2,83	12,37
	41,89	448,24	2,43	14,56
	421,1	4808,93	16,46	-15,04

Skenario	cpu_mean	cpu_max	ram_mean	ram_max
SnortML IPS	21,94	33,01	3271,05	3282,32
Tanpa Snort	3,01	6,37	3015,32	3019,4
Selisih Perubahan (%)	628,9	418,21	8,48	8,71
SQL INJECTION	disk_mean	disk_max	net_mean	net_max
	118,67	17813,73	3,89	34,49
	37,14	314,01	2,12	36,85
	219,52	5572,98	83,49	-6,4

Skenario	cpu_mean	cpu_max	ram_mean	ram_max
SnortML IPS	16,84	26,23	3232,38	3292
Tanpa Snort	1,61	3,18	2963,66	2969,91
Selisih Perubahan (%)	945,96	724,84	9,07	10,85
XSS	disk_mean	disk_max	net_mean	net_max
	875,33	25176,25	2,08	13,79
	41,72	668,9	1,24	11,82
	1998,11	3663,83	67,74	16,67

HARDWARE ANALYSIS

BENIGN

Metric	Tanpa SnortML	SnortML IPS	Selisih	Persentase Selisih (%)
Throughput (kbps)	11.909	6.668	-5.240	-44.006
Average delay (s)	266	148	-117	-44.172
Average jitter (s)	483	263	-220	-45.609
Packet Loss (%)	49.367	46.860	-2.507	-5.079

DoS GOLDENEYE

Metric	Tanpa SnortML	SnortML IPS	Selisih	Presentasi Selisih (%)
Throughput (kbps)	1.320.350	2.730.803	1.410.452	106.824
Average delay (s)	4	2	-2	-51.274
Average jitter (s)	7	3	-4	-51.973
Packet Loss (%)	48.932	47.745	-1.187	-2.426

QOS ANALYSIS



Slow HTTP DoS

Metric	Tanpa SnortML	SnortML IPS	Selisih	Presentas Selisih (%)
Throughput (kbps)	104.979	185.607	80.628	76.805
Average Delay (s)	8	5	-4	-44.532
Average Jitter (s)	14	9	-5	-37.108
Packet Loss (%)	49.074	48.724	-350	-714

BRUTE FORCE

Metric	Tanpa SnortML	SnortML IPS	Selisih	Presentase Selisih (%)
Throughput (kbps)	15.408	12.500	-2.991	-18.872
Average Delay (s)	138	220	82	58.976
Average Jitter (s)	240	420	181	75.413
Packet Loss (%)	45.508	36.718	-8.791	-19.317

SQL Injection

Metric	Tanpa SnortML	SnortML IPS	Selisih	Persentase Selisih (%)
Throughput (kbps)	10.813	7.825	-2.988	-27,634
Average Delay (s)	162	293	132	81.325
Average Jitter (s)	292	567	275	94.203
Packet Loss (%)	45.803	41.146	-4.658	-10.169

QOS ANALYSIS



XSS

Metric	Tanpa SnortML	SnortML IPS	Selisih	Presentas Selisih (%)
Throughput (kbps)	104.979	185.607	80.628	76.805
Average Delay (s)	8	5	-4	-44.532
Average Jitter (s)	14	9	-5	-37.108
Packet Loss (%)	49.074	48.724	-350	-714

CONCLUSION

- This research successfully develops a Machine Learning based Hybrid Intrusion Prevention System (IPS) that enables adaptive and effective cyberattack detection.
- Integrated Snort with anomaly-based ML to detect known and zero-day attacks through layered detection.
- The ML model effectively reduces false positives, achieving accuracy above 85%.
- The system operates in near real-time, with responsive alert visualization on the Wazuh SIEM platform.
- CPU and memory usage remain stable under both benign traffic and attack conditions, with CPU utilization peaking at 73% and memory usage reaching approximately 3000mb.
- The impact on Quality of Service (QoS) is partially achieved, with acceptable throughput under DoS conditions, but increased delay, jitter, and packet loss in application-layer attacks due to ML and deep inspection overhead.

RECOMMENDATION

- Future improvements include a centralized management dashboard for real-time monitoring and incident analysis.
- Using multi-model and multi-classifier ML can improve detection accuracy and robustness against evolving attacks.
- Adaptive multi-action responses with severity-based mitigation can enhance prevention effectiveness.
- LLM integration can assist in log analysis, incident summarization, and response recommendations.
- Deploying the system at the network edge (router/gateway) enables earlier detection and stronger overall network protection.



THANK YOU

So Much